Project Proposal

Customer Intelligence & Retention Platform (CIRP)

DEPI â€¢ AI & Machine Learning Track â€¢ Round 5 â€¢ November 2026

Project: Customer Intelligence & Retention Platform (CIRP)

Objective:

- Predict which customers will stop buying in the next 90 days, and explain why.
- Group customers into behavioural segments with a name and a recommended action.
- Show both results in one Streamlit app, with a risk level and the top reasons for every customer.
- Compare the Random Forest with three deep-learning models and keep the better one on the test split.

Description:

CIRP is a churn prediction and customer segmentation platform for a retail business. It uses the public Online Retail II dataset: 5 quarterly snapshots, 5,253 customers and 22 behavioural features. The 22 features are calculated from the invoice lines (invoice number and date, customer ID, product code and name, quantity and amount); each feature summarises the buying history before the snapshot date. A customer is churned when they place no order in the 90 days after the snapshot. Data is split by time into train (the first three snapshots), validation (June 2011) and test (September 2011) to avoid leakage. A Random Forest predicts churn probability, K-Means groups customers into 6 segments, and the dashboard shows the risk, the top reasons and the segment action. A deep-learning comparison (MLP, FT-Transformer and TabNet) was run and did not replace the Random Forest.

Scope: data preparation and EDA, churn model, segmentation model, explanations, the Streamlit app (local), a deep-learning comparison in an isolated folder, and documentation.

Out of scope for now: cloud deployment, real-time data, CRM integration and sending campaigns.

Tools & Design:

- Data layer: raw transactions -> remove duplicate lines and cancelled invoices -> customer features -> time-based split -> scaling fitted on the training rows -> data package.

- Model layer: Random Forest churn model + K-Means segmentation, with evaluation reports and saved joblib models.
- App layer: Streamlit app loads the saved models and shows churn risk, top factors (feature importance), segment and recommended action.
- Dashboard contract: the app receives a fixed 11-feature preset; cleaning and feature engineering stay in the Python backend.
- Deep-learning layer: separate folder, separate dataset copy, 30 configurations of three networks, and a written comparison report.

Technologies to Use:

- Python, pandas, NumPy and Jupyter Notebook.
- scikit-learn 1.9: Logistic Regression, Decision Tree, Random Forest, K-Means, DBSCAN and StandardScaler.
- joblib for models; matplotlib and seaborn for charts and evaluation reports.
- Streamlit for the app; Git and GitHub for version control (one branch per member, pull requests).
- PyTorch and pytorch-tabnet for the deep-learning comparison; Plotly for the app charts; openpyxl to read the Excel workbook.

Team Members & Responsibilities:

1- **Mohamed Atef Mohamed Abdo** - Data source & preprocessing (Project Leader) - data source, EDA, data dictionary, documentation, system analysis & design and delivery.

2- **Mohamed Ahmed Ebrahim Mohamed** - Data source & preprocessing - cleaning, feature engineering, scaling and the saved preprocessor.

3- **Omar Diaa El-Sayed** - Segmentation - K-Means vs DBSCAN, number of segments, profiles and business names.

4- **Sohaib Mohamed Mohamed Mustafa** - Prediction - churn classifiers, tuning, threshold, explanations and evaluation.

5- **Mohamed Hamada Omar** - Deep Learning - MLP, FT-Transformer and TabNet comparison, validation-only selection, test-access log and the written keep-or-replace decision.

6- **Hussein El-Desouky Ibrahim Othman** - Integration & Deep Learning - model loading, batch scoring, input validation and end-to-end testing, plus deep-learning experiments with Hamada.

Current Results (November 2026)

- Data: 22,372 customer records, 5,253 customers and 61.0% churn; split into 12,178 train, 4,941 validation and 5,253 test rows.
- Churn model: Random Forest at threshold 0.45, with test PR-AUC 0.8224, ROC-AUC 0.8010, recall 95.87%, F1 78.32% and accuracy 69.94%.
- Segmentation: K-Means with k = 6 on nine features; the churn rate per segment ranges from 18.3% (Loyal Champions) to 82.9% (One-Time Lapsed).
- App: five pages (overview, segments, simulator, batch and performance); one score takes under 2 seconds and 5,253 rows under 30 seconds.

Deep Learning Comparison (October - November 2026)

- Three networks were tested in an isolated folder: an MLP, an FT-Transformer and TabNet, with 30 configurations in total.
- Models were selected on the validation split only, and each family was evaluated once on the test split.
- Best validation result: FT-Transformer with PR-AUC 0.8940, against 0.8911 for the Random Forest.
- On test the FT-Transformer reached PR-AUC 0.8082 and recall 93.14%, against 0.8224 and 95.87% for the Random Forest.
- Decision: the Random Forest stays in production; the deep-learning code and report are kept as a documented experiment.

Phase 1 (12 - 25 Oct 2026): Data and baselines

- Mohamed Ahmed - duplicate removal, cancelled invoices, 22 features, five snapshots and the churn label.
- Sohaib - baseline churn models (Logistic Regression, Decision Tree, Random Forest) with Recall, F1 and PR-AUC.
- Omar - first K-Means run and K selection (elbow, silhouette).
- Hussein - Streamlit skeleton, model-app interface and the 11-feature preset.

Deliverables:

- Dataset v1 with five snapshots and the time split.
- Baseline results table and K selection plots.
- Running Streamlit skeleton and sample customers.

Phase 2 (26 Oct - 8 Nov 2026): Models, app and documents

- Sohaib - tune the churn model, choose the threshold on validation and add the explanations.
- Omar - K-Means vs DBSCAN, 6 segments with names, descriptions and actions.
- Hussein - result views, segment page and model integration.
- Hamada and Hussein - deep-learning environment, dataset copy and the first runs.
- Mohamed Atef - requirements, system analysis & design and the interface contract.

Deliverables:

- Tuned model with an evaluation report.
- 6 segment profiles.
- Five app pages connected to the saved models.
- Requirements and design documents.

Phase 3 (9 - 22 Nov 2026): Deep learning and the final test

- Hamada and Hussein - 30 configurations of MLP, FT-Transformer and TabNet, selected on validation.
- Sohaib - final fit on train + validation and one test evaluation per model family.
- Hussein - batch scoring, input validation and end-to-end tests.
- Mohamed Atef - design QA and document updates.

Deliverables:

- Deep-learning report with a keep-or-replace decision.
- Final model files and test metrics.
- Integrated app and test report.

Phase 4 (23 Nov - 4 Dec 2026): Delivery

- Mohamed Atef - final report, README, presentation and GitHub upload.
- Sohaib and Omar - evaluation figures and the data dictionary.
- Hamada and Hussein - deep-learning appendix and the final acceptance checks.

Deliverables:

- Final report and presentation (4 Dec 2026).
- Tagged repository with the frozen data and model files.
- Recorded demo of the five pages.

Full plan, task list and risk register: docs/project_documents/1_Project_Planning_and_Management.docx